

Determine whether a function is even, odd, or neither (symmetry)

Pre-Cal | CK-12: Symmetry

Key Idea

A function is **even** (y-axis symmetry) if $f(-x) = f(x)$, and **odd** (origin symmetry) if $f(-x) = -f(x)$. If neither holds, it is **neither**.

Key Terms

- **Even function** — mirror over the y-axis
- **Odd function** — 180° turn about the origin

Worked steps

Step 1: Replace x with $-x$ and simplify.

Step 2: Compare to $f(x)$ and to $-f(x)$.

Example: $f(x) = x^2$ gives $f(-x) = (-x)^2 = x^2 = f(x)$, so it is **even**.

You Try

Test $f(x) = x^3$ for even/odd.